UNIT III:- SOFTWARE TESTING STRATEGIES

3.1 A Strategic Approach to Software Testing:-

Testing is a set of activities that can be planned in advance and conducted systematically.

A number of software testing strategies have been proposed in the literature. All provide the software developer with a template for testing and all have the following **generic characteristics:**

• Testing begins at the component level2 and works "outward" toward the integration of the entire computer-based system.

• Different testing techniques are appropriate at different points in time.

• Testing is conducted by the developer of the software and (for large projects) an independent test group.

• Testing and debugging are different activities, but debugging must be accommodated in any testing strategy.

- Verification and Validation
- Software Testing Strategies

3.1.1 Verification and Validation:-

Software testing is one element of a broader topic that is often referred to as verification and validation (V&V). Verification refers to the set of activities that ensure that software correctly implements a specific function.

Validation refers to a different set of activities that ensure that the software that has been built is traceable to customer requirements. Boehm [BOE81] states this another way:

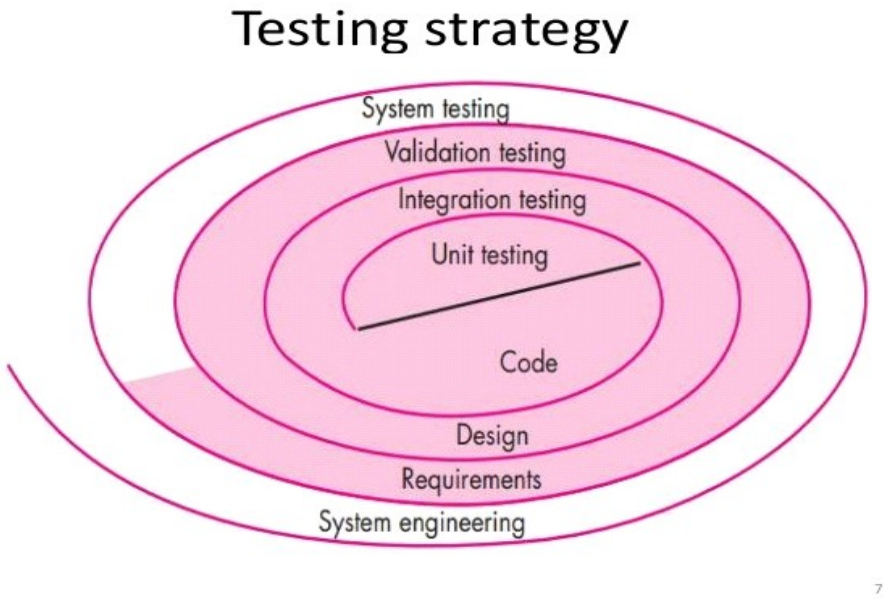
Verification: "Are we building the product right?"
Validation: "Are we building the right product?"

Verification and validation encompasses a wide array of SQA activities that include formal technical reviews, quality and configuration audits, performance monitoring, simulation, feasibility study, documentation

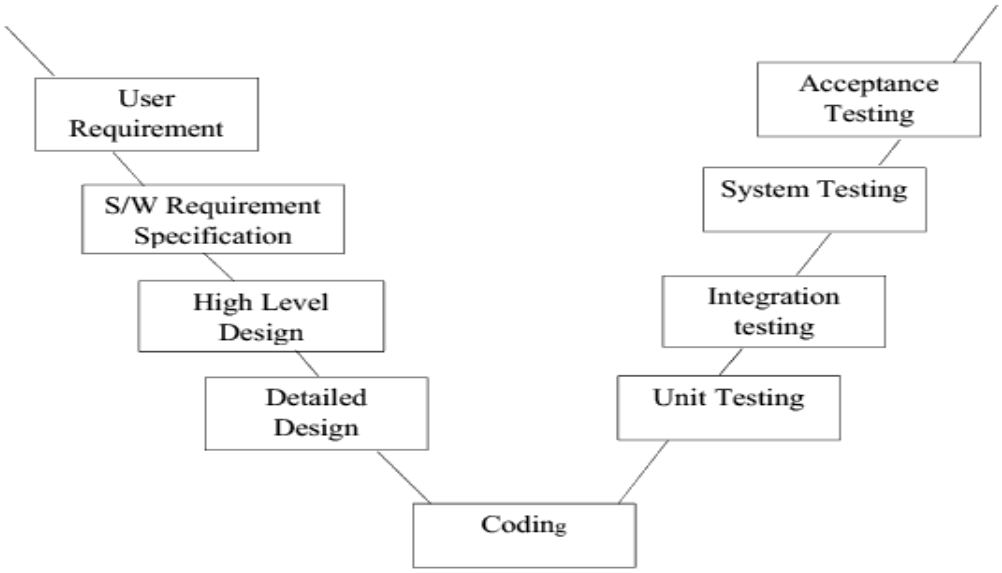
review, database review, algorithm analysis, development testing, qualification testing, and installation testing [WAL89]. Although testing plays an extremely important role in V&V, many other activities are also necessary. Testing does provide the last bastion from which quality can be assessed and, more pragmatically, errors can be uncovered. But testing should not be viewed as a safety net. As they say, "You can't test in quality. If it's not there before you begin testing, it won't be there when you're finished testing." Quality is incorporated into software throughout the process of software engineering. Proper application of methods and tools, effective formal technical reviews, and solid management and measurement all lead to quality that is confirmed during testing.

2)Software Testing Strategy

The software engineering process may be viewed as the spiral model. Initially, system engineering defines the role of software and leads to software requirements analysis, where the information domain, function, behavior, performance, constraints, and validation criteria for software are established. Moving



1). Unit testing begins at the vertex of the spiral and concentrates on each unit (i.e., component) of the software as implemented in source code. Testing progresses by moving outward along the spiral to integration testing, where the focus is on design and the construction of the software architecture. Taking another turn outward on the spiral, we encounter validation testing, where requirements established as part of software requirements analysis are validated against the software that has been constructed. Finally, we arrive at system testing, where the software and other system elements are tested as a whole. To test computer software, we spiral out along streamlines that broaden the scope of testing with each turn.



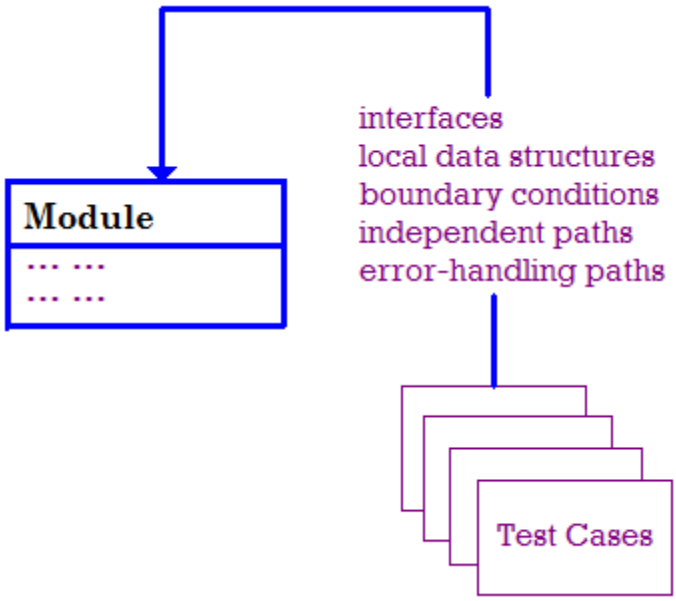
The last high-order testing step falls outside the boundary of software engineering and into the broader context of computer system engineering. Software, once validated, must be combined with other system elements (e.g., hardware, people, databases). System testing verifies that all elements mesh properly and that overall system function/performance is achieved.

3.2 Unit Testing :---

Unit testing focuses verification effort on the smallest unit of software design—the software component or module. Using the component-level design description as a guide, important control paths are tested to uncover errors within the boundary of the module

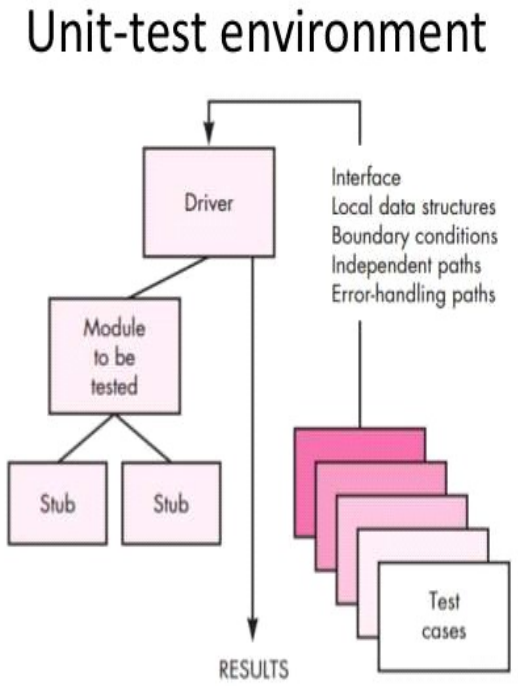
1. Unit Test Considerations
2. Unit Test Procedures

3.2.1. Unit Test Considerations



- The module interface is tested to ensure that information properly flows into and out of the program unit under test.
- The local data structure is examined to ensure that data stored temporarily maintains its integrity during all steps in an algorithm's execution.

- Boundary conditions are tested to ensure that the module operates properly at boundaries established to limit or restrict processing.
- All independent paths (basis paths) through the control structure are exercised to ensure that all statements in a module have been executed at least once.
- And finally, all error handling paths are tested.



3.2.2. Unit Test Procedures

Unit testing is normally considered as an adjunct to the coding step. After source level code has been developed, reviewed, and verified for correspondence to component level design, unit test case design begins. A review of design information provides guidance for establishing test cases that are likely to uncover errors in each of the categories discussed earlier. Each test case should be coupled with a set of expected results. Because a component is not a stand-alone program, driver and/or stub software must be developed for each unit test.

In most applications a ***driver*** is nothing more than a "main program" that accepts test case data, passes such data to the component (to be tested), and prints relevant results.

Stubs serve to replace modules that are subordinate (called by) the component to be tested.

A stub or "dummy subprogram" uses the subordinate module's interface, may do minimal data manipulation, prints verification of entry, and returns control to the module undergoing testing. Drivers and stubs represent overhead.

That is, both are software that must be written (formal design is not commonly applied) but that is not delivered with the final software product. If drivers and stubs are kept simple, actual overhead is relatively low.

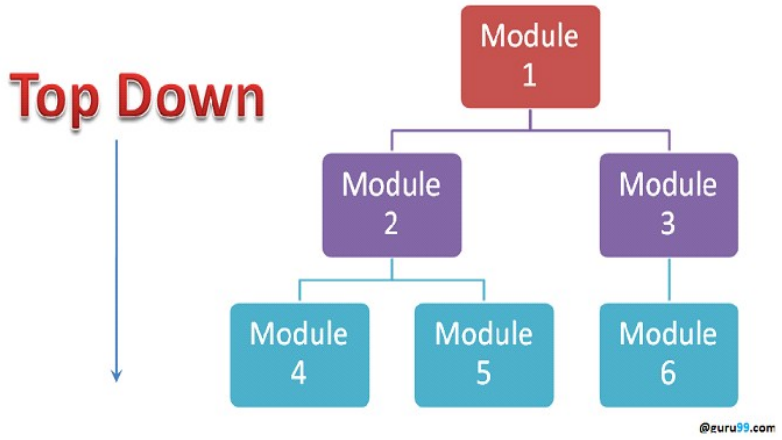
3.3 Integration Testing:--

Integration testing is a systematic technique for constructing the program structure while at the same time conducting tests to uncover errors associated with interfacing. The objective is to take unit tested components and build a program structure that has been dictated by design.

Top-down Integration
Bottom-up Integration
Regression Testing

1) Top-down Integration

Top-down integration testing is an incremental approach to construction of program structure. Modules are integrated by moving downward through the control hierarchy, beginning with the main control module (main program). Modules subordinate (and ultimately subordinate) to the main control module are incorporated into the structure in either a depth-first or breadth-first manner

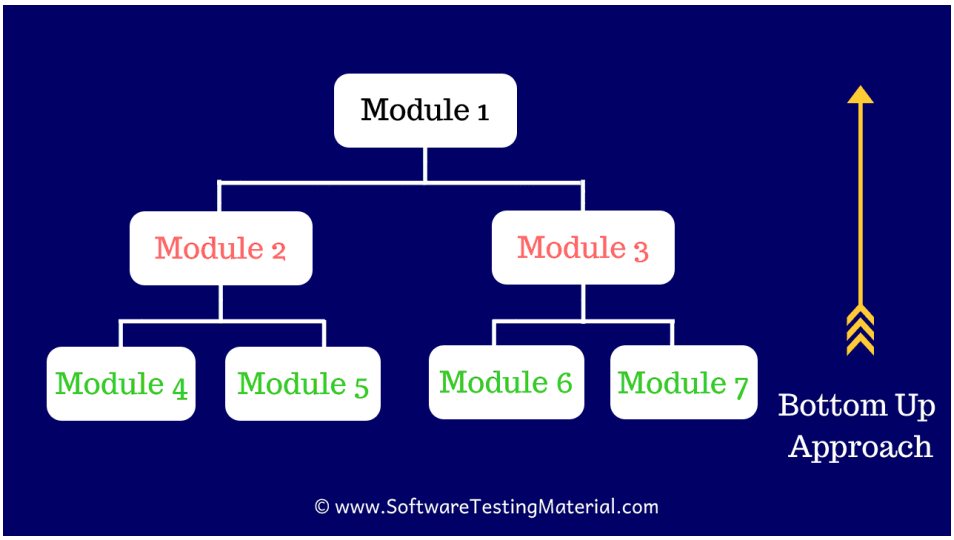


2) Bottom-up Integration

Bottom-up integration testing, as its name implies, begins construction and testing with *atomic modules* (i.e., components at the lowest levels in the program structure). Because components are integrated from the bottom up, processing required for components subordinate to a given level is always available and the need for stubs is eliminated.

A bottom-up integration strategy may be implemented with the following steps:

1. Low-level components are combined into clusters (sometimes called *builds*) that perform a specific software subfunction.
2. A driver (a control program for testing) is written to coordinate test case input and output.
3. The cluster is tested.
4. Drivers are removed and clusters are combined moving upward in the program structure.



3) **Regression Testing**

The regression test suite (the subset of tests to be executed) contains three different

Classes of test cases:

- A representative sample of tests that will exercise all software functions.
- Additional tests that focus on software functions that are likely to be affected by the change.
- Tests that focus on the software components that have been changed. As integration testing proceeds, the number of regression tests can grow quite large.

Therefore, the regression test suite should be designed to include only those tests that address one or more classes of errors in each of the major program functions. It is impractical and inefficient to re-execute every test for every program function once a change has occurred.

Regression testing may be conducted manually, by re-executing a subset of all test cases or using automated *capture/playback tools*. Capture/playback tools enable the software engineer to capture test cases and results for subsequent playback and comparison

3.4 Validation Testing:-

At the culmination of integration testing, software is completely assembled as a package,
interfacing errors have been uncovered and corrected, and a final series of software tests *validation testing*—may begin. Validation can be defined in many ways, but a simple (albeit harsh) definition is that validation succeeds when software functions in a manner that can be reasonably expected by the customer. At this point a battle-hardened software developer might protest: "Who or what is the arbiter of reasonable expectations?"

Validation Test Criteria
Configuration Review
Alpha and Beta Testing

Validation Test Criteria:-

Software validation is achieved through a series of black-box tests that demonstrate conformity with requirements. A test plan outlines the classes of tests to be conducted and a test procedure defines specific test cases that will be used to demonstrate conformity with requirements. Both the plan and procedure are designed to ensure that all functional requirements are satisfied, all behavioral characteristics are achieved, all performance requirements are attained, documentation is correct, and human engineered and other requirements are met (e.g., transportability, compatibility, error recovery, maintainability).
After each validation test case has been conducted, one of two possible conditions exist: (1) The function or performance characteristics conform to specification and are accepted.
(2) a deviation from specification is uncovered and a *deficiency list* is created. Deviation or error discovered at this stage in a project can

rarely be corrected prior to scheduled delivery. It is often necessary to negotiate

Configuration Review

An important element of the validation process is a *configuration review*. The intent of the review is to ensure that all elements of the software configuration have been properly developed, are cataloged, and have the necessary detail to bolster the support phase of the software life cycle.

Alpha and Beta Testing

The *alpha test* is conducted at the developer's site by a customer. The software is used in a natural setting with the developer "looking over the shoulder" of the user and recording errors and usage problems. Alpha tests are conducted in a controlled environment.

The *beta test* is conducted at one or more customer sites by the end-user of the software. Unlike alpha testing, the developer is generally not present. Therefore, the beta test is a "live" application of the software in an environment that cannot be controlled by the developer. The customer records all problems (real or imagined) that are encountered during beta testing and reports these to the developer at regular intervals. As a result of problems reported during beta tests, software engineers make modifications and then prepare for release of the software product to the entire customer base.

3.5 System Testing:-

At the beginning of this book, we stressed the fact that software is only one element of a larger computer-based system. Ultimately, software is incorporated with other system elements (e.g., hardware, people, information), and a series of system integration and validation tests are conducted.

A classic system testing problem is "finger-pointing." This occurs when an error is uncovered, and each system element developer blames the

other for the problem. Rather than indulging in such nonsense, the software engineer should anticipate potential interfacing problems and
(1) Design error-handling paths that test all information coming from other elements of the system.
(2) Conduct a series of tests that simulate bad data or other potential errors at the software interface.
(3) Record the results of tests to use as "evidence" if finger-pointing does occur.
(4) Participate in planning and design of system tests to ensure that software is adequately tested.

Recovery Testing
Security Testing
Stress Testing
Performance Testing

Recovery Testing:-
Recovery testing is a system test that forces the software to fail in a variety of ways and verifies that recovery is properly performed. If recovery is automatic (performed by the system itself), reinitialization, check pointing mechanisms, data recovery, and restart are evaluated for correctness. If recovery requires human intervention, the mean-time-to-repair (MTTR) is evaluated to determine whether it is within acceptable limits

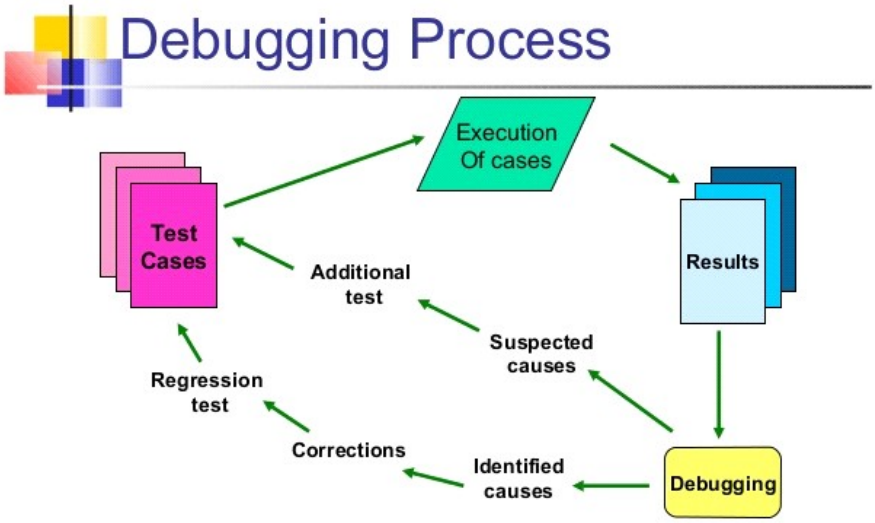
Security Testing:-
Any computer-based system that manages sensitive information or causes actions that can improperly harm (or benefit) individuals is a target for improper or illegal penetration. Penetration spans a broad range of activities: hackers who attempt to penetrate systems for sport; disgruntled employees who attempt to penetrate for revenge; dishonest individuals who attempt to penetrate for illicit personal gain.

Security testing attempts to verify that protection mechanisms built into a system will, in fact, protect it from improper penetration. The system's security must, of course, be tested for invulnerability from frontal attack—but must also be tested for invulnerability from flank or rear attack

Stress Testing:-

Stress testing executes a system in a manner that demands resources in abnormal quantity, frequency, or volume.
For example,
(1) Special tests may be designed that generate ten interrupts per second, when one or two is the average rate,
(2) Input data rates may be increased by an order of magnitude to determine how input functions will respond,
(3) Test cases that require maximum memory or other resources are executed,
(4) Test cases that may cause thrashing in a virtual operating system are designed,
(5) Test cases that may cause excessive hunting for disk-resident data are created. Essentially, the tester attempts to break the program.
A variation of stress testing is a technique called sensitivity testing. In some situations (the most common occur in mathematical algorithms), a very small range of data contained within the bounds of valid data for a program may cause extreme and even erroneous processing or profound performance degradation. Sensitivity testing attempts to uncover data combinations within valid input classes that may cause instability or improper processing.

Performance Testing:-



For real-time and embedded systems, software that provides required function but does not conform to performance requirements is unacceptable. *Performance testing* is designed to test the run-time performance of software within the context of an integrated system. Performance testing occurs throughout all steps in the testing process. Even at the unit level, the performance of an individual module may be assessed as white-box tests are conducted. However, it is not until all system elements are fully integrated that the true performance of a system can be ascertained.

Performance tests are often coupled with stress testing and usually require both hardware and software instrumentation. That is, it is often necessary to measure resource utilization (e.g., processor cycles) in an exacting fashion. External instrumentation can monitor execution intervals, log events (e.g., interrupts) as they occur, and sample machine states on a regular basis. By instrumenting a system, the tester can uncover situations that lead to degradation and possible system failure.

3.6 The Art of Debugging

Software testing is a process that can be systematically planned and specified. Test case design can be conducted, a strategy can be defined, and results can be evaluated against prescribed expectations.

Debugging occurs as a consequence of successful testing. That is, when a test case uncovers an error, debugging is the process that results in the removal of the error.

Although debugging can and should be an orderly process, it is still very much an art.

A software engineer, evaluating the results of a test, is often confronted with a "symptomatic" indication of a software problem. That is, the external manifestation of the error and the internal cause of the error may have no obvious relationship to one another. The poorly understood mental process that connects a symptom to a cause is debugging

The Debugging Process
Psychological Considerations
Debugging Approaches

The Debugging Process

Debugging is not testing but always occurs as a consequence of testing. The debugging process begins with the execution of a test case. Results are assessed and a lack of correspondence between expected and actual performance is encountered.

The debugging process will always have one of two outcomes:

(1) the cause will be found and corrected.

(2) the cause will not be found. In the latter case, the person performing debugging may suspect a cause, design a test case to help validate that suspicion, and work toward error correction in an iterative fashion.

Why is debugging so difficult?
characteristics of bugs provide some clues:
1. The symptom and the cause may be geographically remote. That is, the symptom may appear in one part of a program, while the cause may actually be located at a site that is far removed. Highly coupled program structures exacerbate this situation.
2. The symptom may disappear (temporarily) when another error is corrected.
3. The symptom may actually be caused by non errors (e.g., round-off inaccuracies).
4. The symptom may be caused by human error that is not easily traced.
5. The symptom may be a result of timing problems, rather than processing problems.
6. It may be difficult to accurately reproduce input conditions (e.g., a real-time application in which input ordering is indeterminate).
7. The symptom may be intermittent. This is particularly common in embedded systems that couple hardware and software inextricably.
8. The symptom may be due to causes that are distributed across a number of tasks running on different processors.

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